```
(3.40%)
      rule_tachycardia: 302
      rule_bradycardia: 43
      rule_low_spo2: 161
      rule_sleep_steps: 81
      rule_sleep_hr: 2

[Prophet Residuals] Merging anomaly flags...
   ✅ Prophet anomalies merged: 2580

[DBSCAN] Merging outlier cluster flags...
   ✅ DBSCAN outliers merged: 965

✅ Combined anomaly score computed.
   Total flagged: 3551 (20.55%)
   Severity breakdown:
severity
normal    13729
low        3037
medium      446
high         68

💾 Anomaly results → outputs/anomaly_results.csv
📊 Heart rate anomaly plot → outputs/milestone3_heartrate.png
📊 Sleep anomaly plot → outputs/milestone3_sleep.png
📊 Steps & SpO2 plot → outputs/milestone3_steps_spo2.png
📊 Anomaly summary plot → outputs/milestone3_summary.png

✅ Milestone 3 Complete!

==============================
✅ ALL MODULES COMPLETED!
==============================
PS D:\info>
```

FOLDERS

INFO

outputs

- anomaly_results.csv
- cleaned_data.csv
- feature_clustered.csv
- feature_matrix.csv
- milestone1_preview....
- milestone2_clusters....
- milestone3_heartrate...
- milestone3_sleep.png
- milestone3_steps_sp...
- milestone3_summary...
- prophet_heart_rate_...
- prophet_spo2_pct.png
- prophet_steps.png

- fitness_data_raw.csv
- milestone1_preproces...
- milestone2_modeling....
- milestone3_anomaly.py
- run_all.py

OUTLINE

TIMELINE

anomaly_results.csv

outputs > anomaly_results.csv > data

```csv
timestamp,heart_rate_bpm,steps,sleeping,spo2_pct,calories_burned,is_anomaly,rule_tachycardia,rule_bradycardia,rule_low_spo2,rule_sleep_steps,rule_sleep_hr,rule_anomaly,prophet_anomaly,dbscan_anomaly,anomaly_score,final_anomaly,se
2024-01-01 00:00:00,62.3,0.0,1.0,97.4,3.32,0.0,0,0,0,0,0,0,0,1,1,1,low
2024-01-01 00:05:00,65.1,0.0,1.0,97.9,2.98,0.0,0,0,0,0,0,0,0,1,1,1,low
2024-01-01 00:10:00,58.1,0.0,1.0,96.5,2.09,0.0,0,0,0,0,0,0,0,1,1,1,low
2024-01-01 00:15:00,54.0,0.0,1.0,96.8,3.45,0.0,0,0,0,0,0,0,0,1,1,1,low
2024-01-01 00:20:00,56.4,0.0,1.0,97.6,2.22,0.0,0,0,0,0,0,0,0,1,1,1,low
2024-01-01 00:25:00,58.4,0.0,1.0,97.4,2.48,0.0,0,0,0,0,0,0,0,1,1,1,low
2024-01-01 00:30:00,58.4,0.0,1.0,96.9,3.17,0.0,0,0,0,0,0,0,0,1,1,1,low
2024-01-01 00:35:00,60.5,0.0,1.0,97.6,2.99,0.0,0,0,0,0,0,0,0,1,1,1,low
2024-01-01 00:40:00,53.5,0.0,1.0,98.0,2.98,0.0,0,0,0,0,0,0,0,1,1,1,low
2024-01-01 00:45:00,54.8,0.0,1.0,97.8,3.28,0.0,0,0,0,0,0,0,0,1,1,1,low
2024-01-01 00:50:00,56.1,0.0,1.0,97.3,2.9,0.0,0,0,0,0,0,0,0,1,1,1,low
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2024-01-01 01:00:00,61.8,0.0,1.0,97.2,3.15,0.0,0,0,0,0,0,0,0,0,0,0,normal
2024-01-01 01:05:00,47.1,0.0,1.0,97.9,2.42,0.0,0,0,0,0,0,0,0,0,0,0,normal
2024-01-01 01:10:00,56.3,0.0,1.0,97.7,2.81,0.0,0,0,0,0,0,0,0,0,0,0,normal
2024-01-01 01:15:00,55.3,0.0,1.0,97.2,3.2,0.0,0,0,0,0,0,0,0,0,0,0,normal
2024-01-01 01:20:00,57.9,0.0,1.0,98.0,2.57,0.0,0,0,0,0,0,0,0,0,0,0,normal
2024-01-01 01:25:00,58.8,0.0,1.0,97.6,2.96,0.0,0,0,0,0,0,0,0,0,0,0,normal
2024-01-01 01:30:00,57.2,0.0,1.0,97.1,2.75,0.0,0,0,0,0,0,0,0,0,0,0,normal
2024-01-01 01:35:00,59.0,0.0,1.0,97.5,1.99,0.0,0,0,0,0,0,0,0,0,0,0,normal
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2024-01-01 01:50:00,59.5,0.0,1.0,98.6,2.52,0.0,0,0,0,0,0,0,0,0,0,0,normal
2024-01-01 01:55:00,52.5,0.0,1.0,97.5,2.06,0.0,0,0,0,0,0,0,0,0,0,0,normal
2024-01-01 02:00:00,53.0,0.0,1.0,97.3,3.15,0.0,0,0,0,0,0,0,0,0,0,0,normal
2024-01-01 02:05:00,52.7,0.0,1.0,97.6,2.71,0.0,0,0,0,0,0,0,0,0,0,0,normal
2024-01-01 02:10:00,60.6,0.0,1.0,97.6,3.13,0.0,0,0,0,0,0,0,0,0,0,0,normal
2024-01-01 02:15:00,59.9,0.0,1.0,97.1,3.89,0.0,0,0,0,0,0,0,0,0,0,0,normal
2024-01-01 02:20:00,52.9,0.0,1.0,97.9,3.28,0.0,0,0,0,0,0,0,0,0,0,0,normal
2024-01-01 02:25:00,60.2,0.0,1.0,98.4,2.94,0.0,0,0,0,0,0,0,0,0,0,0,normal
2024-01-01 02:30:00,58.9,0.0,1.0,97.7,3.02,0.0,0,0,0,0,0,0,0,0,0,0,normal
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2024-01-01 02:40:00,54.8,0.0,1.0,97.7,2.57,0.0,0,0,0,0,0,0,0,0,0,0,normal
2024-01-01 02:45:00,57.6,0.0,1.0,97.5,2.57,0.0,0,0,0,0,0,0,0,0,0,0,normal
2024-01-01 02:50:00,59.4,0.0,1.0,96.9,3.03,0.0,0,0,0,0,0,0,0,0,0,0,normal
2024-01-01 02:55:00,59.1,0.0,1.0,96.9,3.09,0.0,0,0,0,0,0,0,0,0,0,0,normal
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2024-01-01 03:05:00,61.8,0.0,1.0,99.4,3.29,0.0,0,0,0,0,0,0,0,0,0,0,normal
2024-01-01 03:10:00,56.4,0.0,1.0,97.9,2.45,0.0,0,0,0,0,0,0,0,0,0,0,normal
2024-01-01 03:15:00,64.8,0.0,1.0,96.6,3.71,0.0,0,0,0,0,0,0,0,0,0,0,normal
2024-01-01 03:20:00,54.5,0.0,1.0,98.0,2.84,0.0,0,0,0,0,0,0,0,0,0,0,normal
2024-01-01 03:25:00,54.1,0.0,1.0,97.6,2.78,0.0,0,0,0,0,0,0,0,0,0,0,normal
2024-01-01 03:30:00,51.2,0.0,1.0,97.6,2.16,0.0,0,0,0,0,0,0,0,0,0,0,normal
2024-01-01 03:35:00,59.5,0.0,1.0,97.9,2.4,0.0,0,0,0,0,0,0,0,0,0,0,normal
2024-01-01 03:40:00,66.4,0.0,1.0,97.7,2.62,0.0,0,0,0,0,0,0,0,0,0,0,normal
2024-01-01 03:45:00,65.8,0.0,1.0,98.0,2.56,0.0,0,0,0,0,0,0,0,0,0,0,normal
2024-01-01 03:50:00,59.7,0.0,1.0,97.9,2.53,0.0,0,0,0,0,0,0,0,0,0,0,normal
2024-01-01 03:55:00,54.8,0.0,1.0,96.9,3.67,0.0,0,0,0,0,0,0,0,0,0,0,normal
2024-01-01 04:00:00,63.8,0.0,1.0,96.8,3.77,0.0,0,0,0,0,0,0,0,0,0,0,normal
2024-01-01 04:05:00,58.2,0.0,1.0,97.2,2.97,0.0,0,0,0,0,0,0,0,0,0,0,normal
2024-01-01 04:10:00,61.4,0.0,1.0,96.9,4.28,0.0,0,0,0,0,0,0,0,0,0,0,normal
2024-01-01 04:15:00,58.4,0.0,1.0,97.2,2.85,0.0,0,0,0,0,0,0,0,0,0,0,normal
2024-01-01 04:20:00,54.6,0.0,1.0,97.9,2.96,0.0,0,0,0,0,0,0,0,0,0,0,normal
2024-01-01 04:25:00,60.4,0.0,1.0,97.8,2.65,0.0,0,0,0,0,0,0,0,0,0,0,normal
```
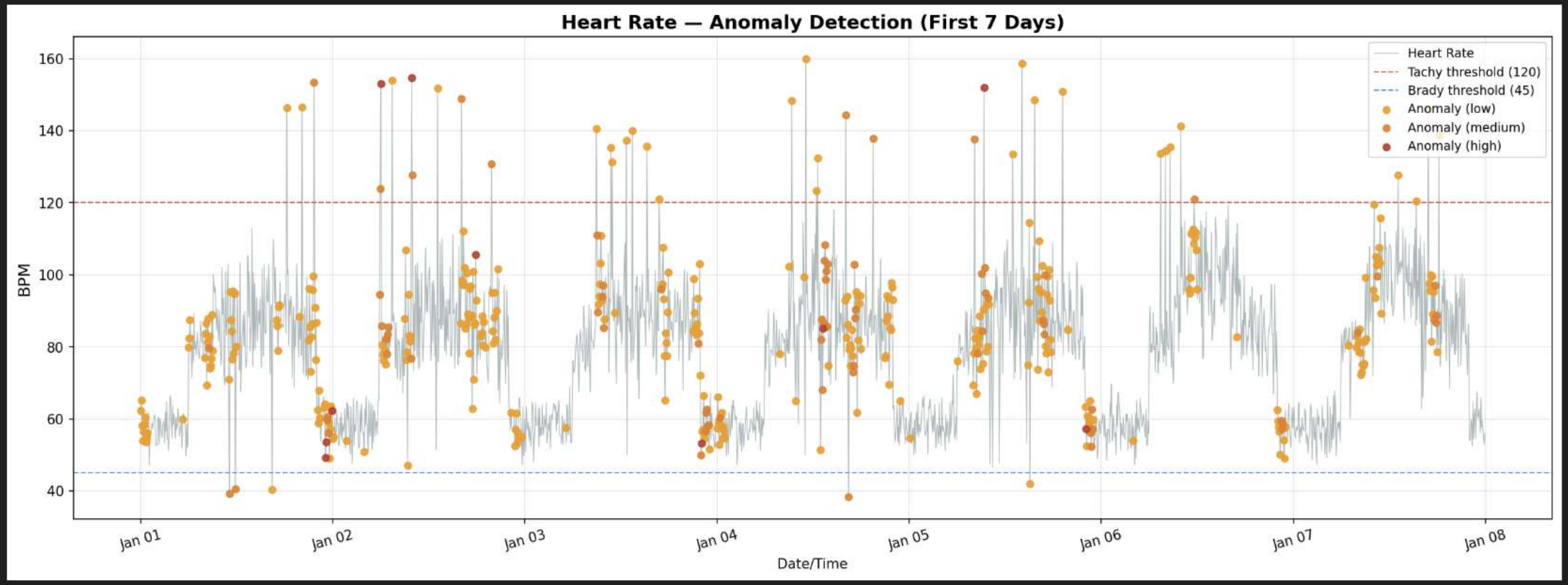
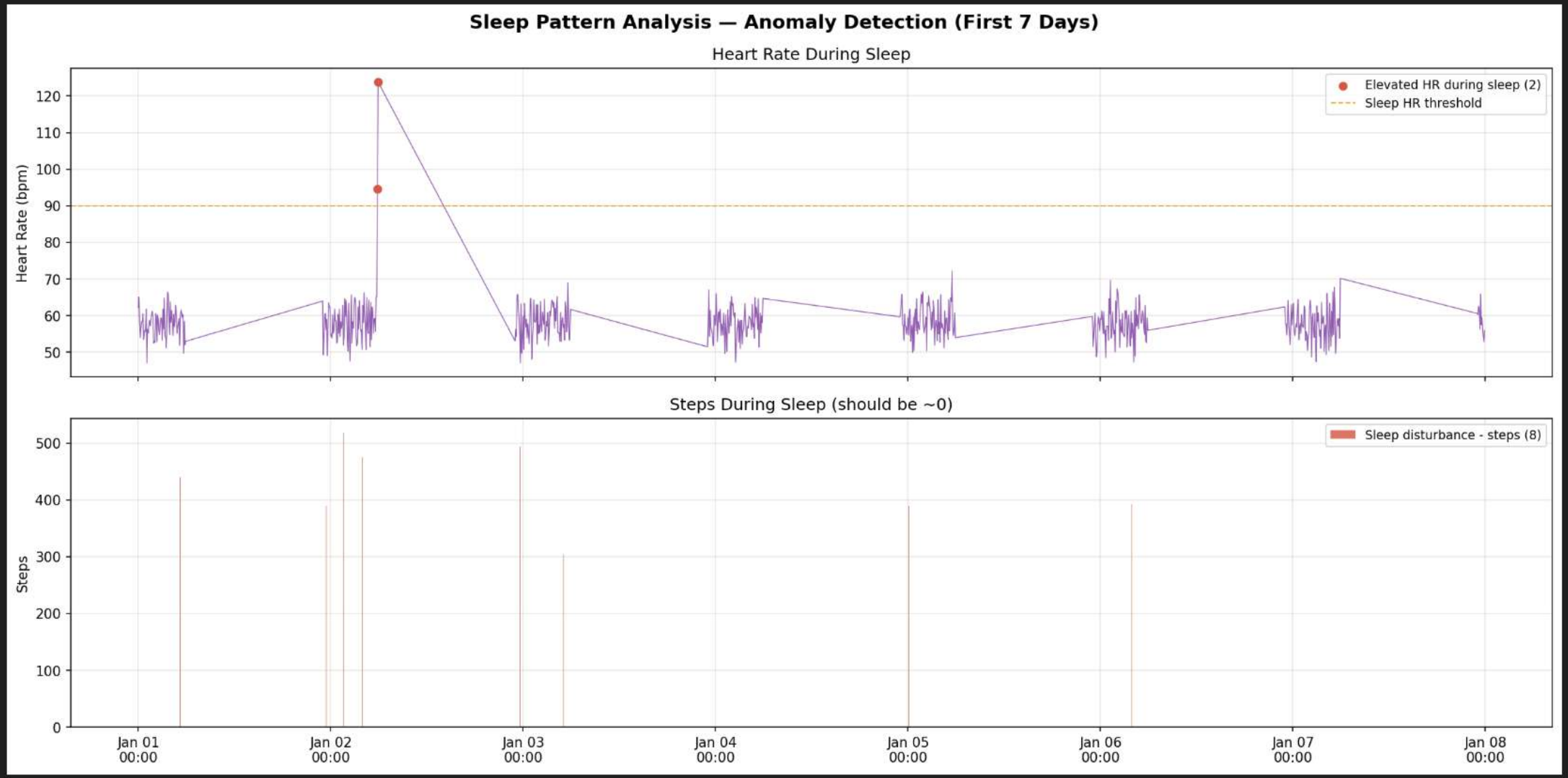

milestone3_heartrate.png

outputs > milestone3_heartrate.png



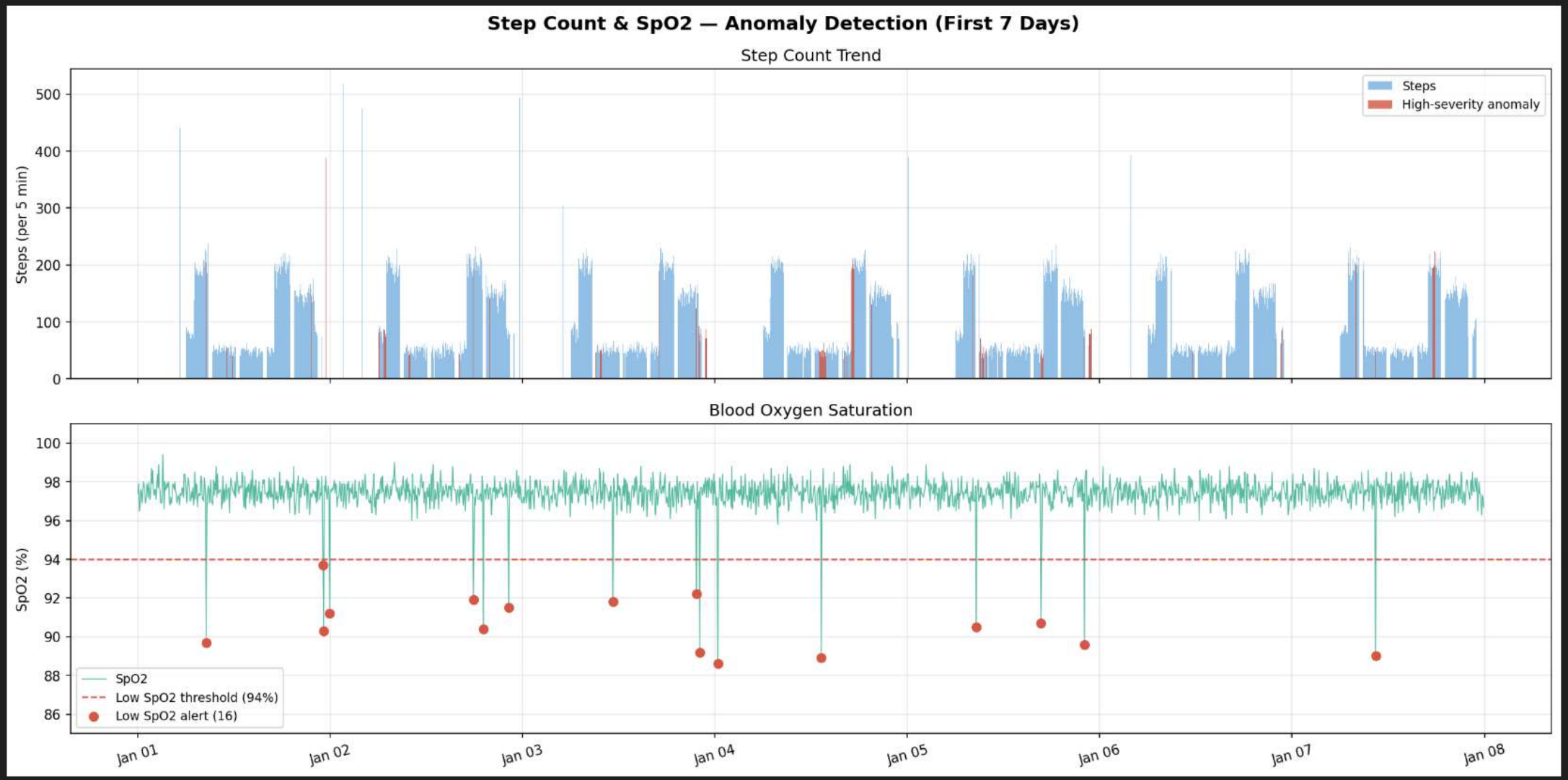
milestone3_sleep.png

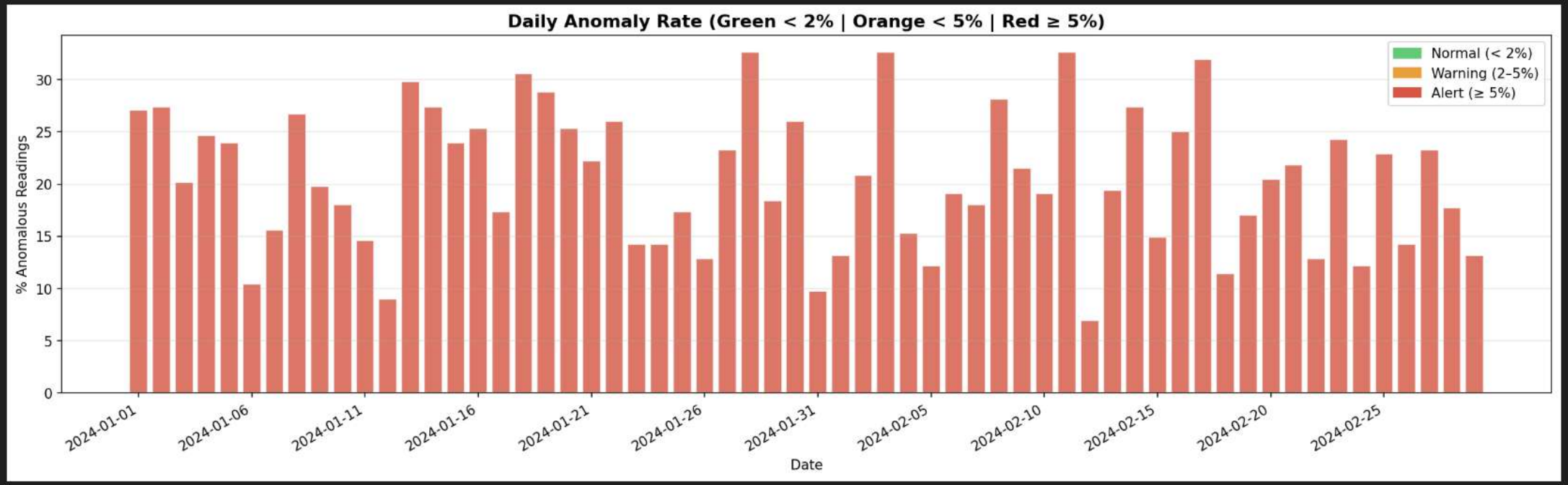
outputs > milestone3_sleep.png



milestone3_steps_spo2.png

outputs > milestone3_steps_spo2.png



milestone3_summary.png

outputs > milestone3_summary.png